



Urchinlike hex-WO₃ microspheres: Hydrothermal synthesis and gas-sensing properties

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ABSTRACT

Novel urchinlike hexagonal WO₃ microspheres have been fabricated by a facile hydrothermal method in the presence of K₂SO₄. Morphology analysis reveals that the as-synthesized WO₃ microstructures are assembled by numerous one-dimensional nanorods. Based on comparative studies, a possible formation mechanism is proposed in detail. What's more, gas-sensing measurement indicates that the well-defined urchinlike WO₃ microspheres exhibit excellent gas sensing properties.

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1. Introduction

Among multitudinous transition metal oxides, nanostructured tungsten trioxide (WO₃) and its derivatives have attracted extensive attention due to their various crystal types, large surface areas, anisotropic tunnel structure [1], and unique physic-chemical properties as well as corresponding applications in electrochromic device [2], photocatalysts [3] and gas sensors [4,5]. So far, much effort has been devoted to synthesizing advanced WO₃ architectures. Of diverse implementation techniques [6–9], the hydrothermal approach [10–12] stands out as a promising route owing to energy conservation, low temperature and controllable particle size. For example, in the presence of plentiful K₂SO₄, Song et al. [13] shaped the WO₃ into one-dimensional nanowires via a simple hydrothermal method. Meanwhile in hydrothermal condition containing C₂H₂O₄, Gu et al. [14] reported the controlled synthesis of urchinlike and ribbonlike WO₃ by adding Rb₂SO₄ and K₂SO₄, respectively. Due to complicated hydrothermal conditions, there may be a thousand morphologies in a thousand autoclaves. Both of the two research groups focus on preparation of various morphologies through adding alkali metal sulfates instead of their applications. So, it is fascinating and significative to fabricate superior WO₃ architectures with the simple process and further put them into application, such as fabricating gas sensor.

Herein, we have reported a facile hydrothermal approach to synthesize urchinlike hexagonal WO₃ (hex-WO₃) microspheres with K₂SO₄ as surfactant in acidized solution. Based on comparative studies,

the growth mechanism is discussed in detail. Moreover, to demonstrate the potential applications, the as-prepared powders are used to fabricate gas sensor which exhibits excellent gas-sensing performance towards ethanol.

2. Experimental

Seven mmol Na₂WO₄ was dissolved in 40 ml deionized water under vigorous magnetic stirring. Then 3.5 mmol K₂SO₄ was added into the solution with sequentially stirring for 60 min. The clear solution was slowly acidified to a pH range of 1.5–2 using dropwise HCl under continuous stirring to form a pale yellow suspension. The newly formed suspension was transferred into a Teflon-lined 50 ml stainless steel autoclave and kept at 180 °C for 24 h, then cooled to room temperature naturally. The products were filtered and washed with ethanol and deionized water several times. Finally, white precipitate was obtained by centrifugation and following drying at 60 °C.

The instrument and process of fabricating a gas sensor were similar to that of our previous work [15]. Gas-sensing properties towards ethanol were measured using a static system controlled by a computer under current laboratory conditions. Certain amount of ethanol liquid was measured by a needle and then evaporated into well-defined concentration of ethanol gas in the reaction chamber. Gas response in this paper is defined as V_g/V_a, where V_g and V_a are the test voltage measured in ethanol gas and in air, respectively. And the response (recovery) time is defined as the time taken by the sensor to achieve 90% of the total voltage change in the case of gas-in (gas-out).

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3. Results and discussion

XRD pattern of the as-synthesized product is shown in Fig. 1. All the diffraction peaks can be well indexed to pure hexagonal WO_3 , compared with JCPDS NO. 75-2187. On the other hand, the intense and sharp peaks indicate the good crystallinity of the hydrothermal product. It is obvious that the relative intensity of the (0 0 1) peak is higher than that of the standard level, implying that the preferential growth of the hex- WO_3 was along the c -axis.

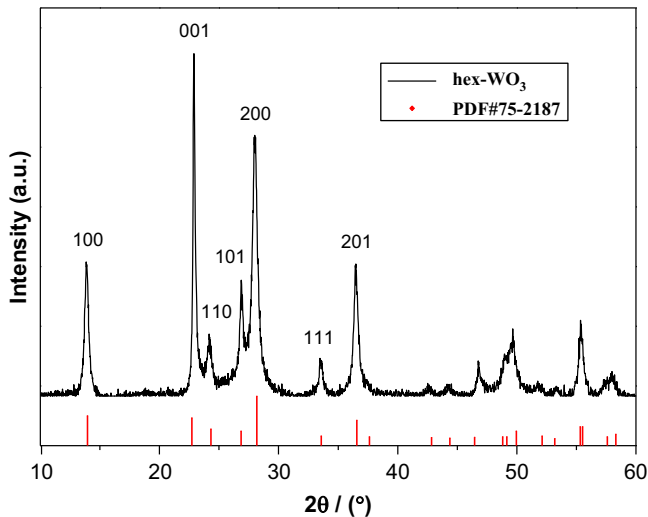


Fig. 1. XRD pattern of the sample.

The morphology and microstructures of the obtained hex- WO_3 were investigated by SEM and TEM. From the panoramic view showed in Fig. 2a, monodisperse hierarchical urchinlike architectures with dimensions of 2–4 μm can be found. When combined with magnified SEM image (Fig. 2b), one can notice that the surfaces of these microspheres are composed of a crowd of well-defined individual nanorods, which further measured ~ 15 –25 nm in diameter and up to 1 μm in length from Fig. 2c. Namely, the aspect ratio was approximately 40 to 67. It also can be observed in Fig. 2c that the outward extended nanorods radiate towards the dense core with slight position shift, agreeing well with the TEM image in low magnification (inset). The HRTEM and the corresponding fast Fourier transform (FFT, inset) images (Fig. 2d) confirm that the hex- WO_3 nanorod was monocrystal, growing along the [0 0 1] direction.

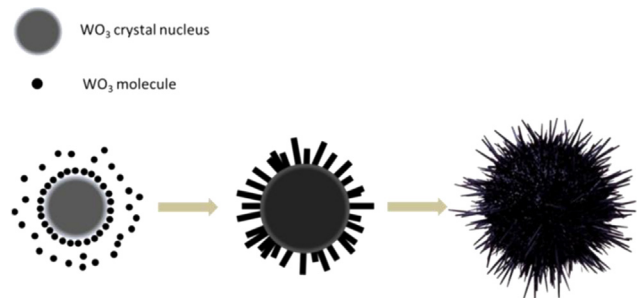


Fig. 3. Schematic illustrations of the possible formation processes of the urchinlike hex- WO_3 microspheres.

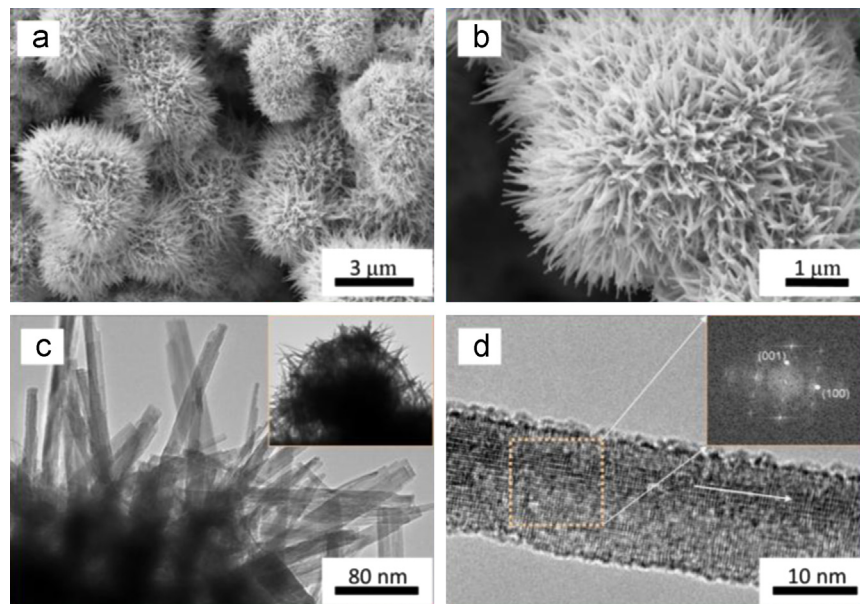


Fig. 2. (a) low magnification and (b) high magnification SEM images, (c) TEM (inset) and high magnification TEM images, (d) HRTEM image and corresponding FFT pattern (inset) of the urchinlike hex- WO_3 microspheres.

Table 1

Investigation of three similar methods but dissimilar outcomes.

Sample	Processes	$n(\text{W}):n(\text{K})^a$	pH	T ($^{\circ}\text{C}$)	t (h)	Morphology
Song et al.[18]	$\text{Na}_2\text{WO}_4 \cdot 2\text{H}_2\text{O} \rightarrow \text{HCl} \rightarrow \text{K}_2\text{SO}_4$	1:151.4	Acidic	180	12	Nanowires
Gu et al.[17]	$\text{Na}_2\text{WO}_4 \cdot 2\text{H}_2\text{O} \rightarrow \text{HCl} \rightarrow \text{H}_2\text{C}_2\text{O}_4 \rightarrow \text{K}_2\text{SO}_4$	2.15:1	1–1.2	180	2–72	Ribbonlike
This work	$\text{Na}_2\text{WO}_4 \cdot 2\text{H}_2\text{O} \rightarrow \text{K}_2\text{SO}_4 \rightarrow \text{HCl}$	1:1	1.5–2.0	180	24	Urchinlike

^a $n(\text{W}):n(\text{K})$ is the molar mass ratio between tungsten and potassium.

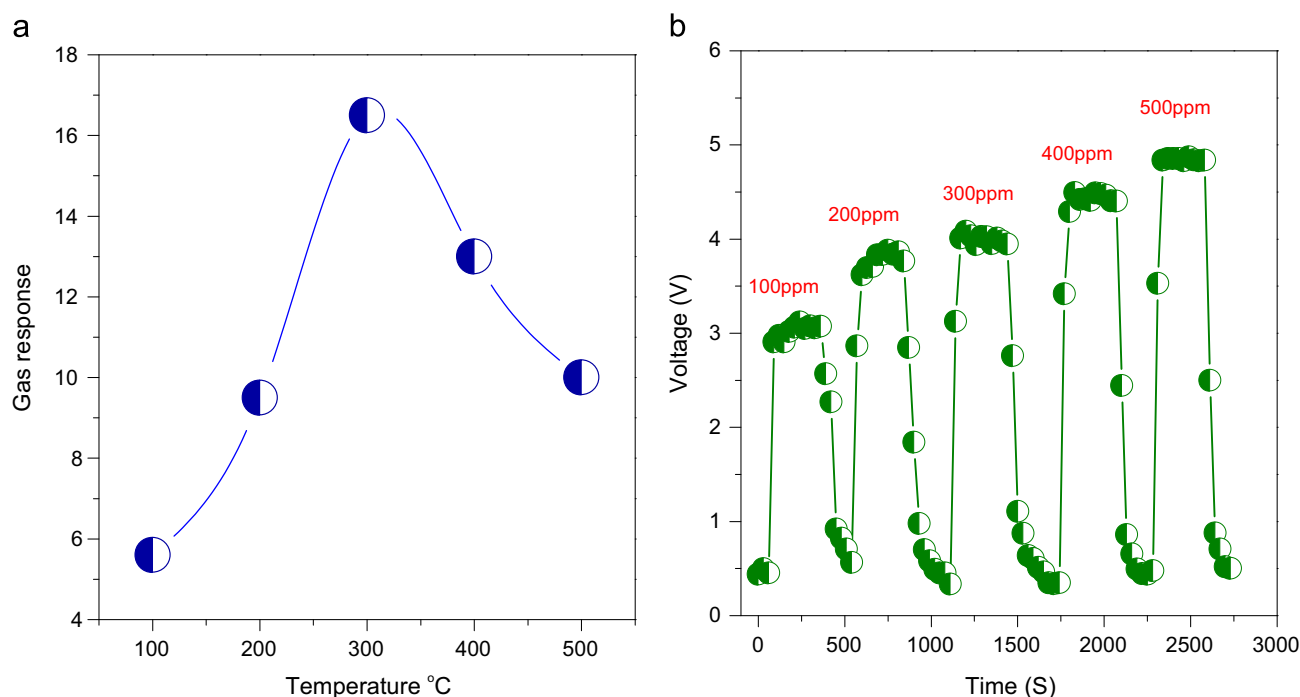


Fig. 4. (a) Response of the sensor to the ethanol at temperatures from 100 to 450 °C. (b) Response of the sensor to different concentrations of ethanol.

Table 1 provides some valuable information of similar results [13,14] to understand the morphology evolution. In essence, although different in $n(\text{W}):n(\text{K})$, all the basic units of the three cases can be ascribed to nanowires, since ribbonlike WO_3 evolved from the oriented attachment of primary nanowires while the nanorods of urchinlike WO_3 could be regarded as grown nanowires. As a typical directing agent, more K_2SO_4 means larger capping scale, this is, more efficient directing capacity [16], which corresponds to the formations of nanobelt, nanorod, and nanowire with the increasing proportion of K_2SO_4 . Thus, with the combined contribution of other experimental factors (including the sequence of adding K_2SO_4 and HCl , reaction time, hydrothermal temperature, and so on), the presence of an appropriate amount of K_2SO_4 plays an important role in the formation of the hex- WO_3 nanorods. Based on the discussion, a possible formation process is demonstrated in Fig. 3. The formation of urchinlike hex- WO_3 in hydrothermal solution occurred in two steps: nucleation and subsequent growth. In terms of the first step, the acidic precipitation H_2WO_4 was formed after HCl added, and then decomposed into large initial WO_3 nuclei with the help of sufficient energy provided by the hydrothermal system. Then the secondary nucleation occurred preferentially at the surface defect sites [17] of the initial nuclei. Meanwhile, the K_2SO_4 adsorbed selectively onto specific faces of the freshly formed secondary crystals, reduced the surface energy and thereby inhibited the growth of these faces, giving rise to the formation of 1D nanorods along [0 0 1] direction. Finally, the urchinlike hex- WO_3 microspheres consisting of numerous high aspect-ratio nanorods were formed.

The unique urchinlike microstructure may play an important role in tailoring the properties of the metal oxides, in particular the gas-sensing behavior. To test the potential application, we first present the gas response to the ethanol at various operating temperatures under a certain gas concentration of 400 ppm in Fig. 4a. Evidently, the sensor exhibited good gas-sensing performance, especially at 300 °C, the gas response of which was estimated to be 17. Fig. 4b shows the measurable and unsaturated sensitive ability in ethanol of increasing concentrations at 300 °C. In case of 400 ppm ethanol, the response and recover times are about 50 s and 70 s, respectively. A comparison (Table S1) of the

ethanol sensing of urchinlike hex- WO_3 we prepared with other sensor materials [5,15,18–21] indicate that the as-prepared urchinlike hex- WO_3 hierarchical architectures exhibit excellent integrated gas sensing performances and are promising candidates for gas sensing application.

4. Conclusions

Novel urchinlike hex- WO_3 microstructures assembled by numerous 1D-nanorods were synthesized through a facile hydrothermal method in the presence of K_2SO_4 . A comparison indicates that appropriate amount of K_2SO_4 plays an important role in determining the ultimate morphologies of WO_3 , which will be further systematically analyzed in the future. The gas-sensing measurements reveal that the WO_3 with high aspect ratio represents an important step forward in exploring novel gas sensors for future practical applications.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at <http://dx.doi.org/10.1016/j.matlet.2015.01.019>.

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